

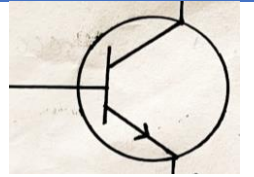
HAND BOOK OF ELECTRONICS

GUPTA & KUMAR

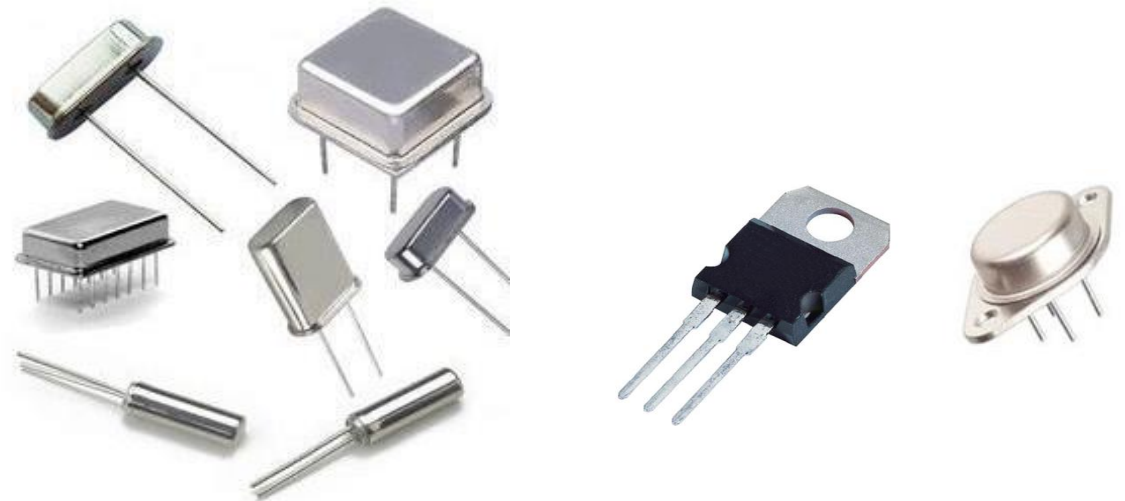
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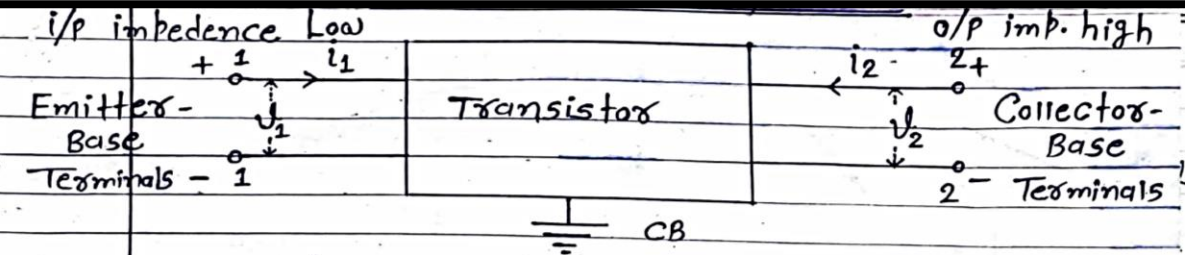
Paper 504



**Unit 2 Basic Transistor Amplifier
& Multistage Transistor Amplifier**



Jatin Patel Sir (MASC, Mehsana, B.Sc. College)



- સારકિટ સારકિટ = i_1, v_2
- સારકિટ સારકિટ = v_2, i_2

$$v_1 = f(i_1, v_2) \text{ and } i_2 = f(i_1, v_2)$$

$$\therefore dv_1 = \left(\frac{\partial v_1}{\partial i_1} \right) di_1 + \left(\frac{\partial v_1}{\partial v_2} \right) dv_2$$

$$di_2 = \left(\frac{\partial i_2}{\partial i_1} \right) di_1 + \left(\frac{\partial i_2}{\partial v_2} \right) dv_2$$

$$\frac{v_1}{i_1} = h_{11} = \text{short circuit input impedance, } v_2 = 0$$

$$= h_i; \text{ Unit} = \Omega (\text{ohm})$$

$$\frac{v_1}{v_2} = h_{12} = \text{open circuit reverse voltage ratio}$$

$$= h_r; i_1 = 0$$

$$\frac{i_2}{i_1} = h_{21} = \text{s.c. forward current ratio, } v_2 = 0$$

$$= h_f$$

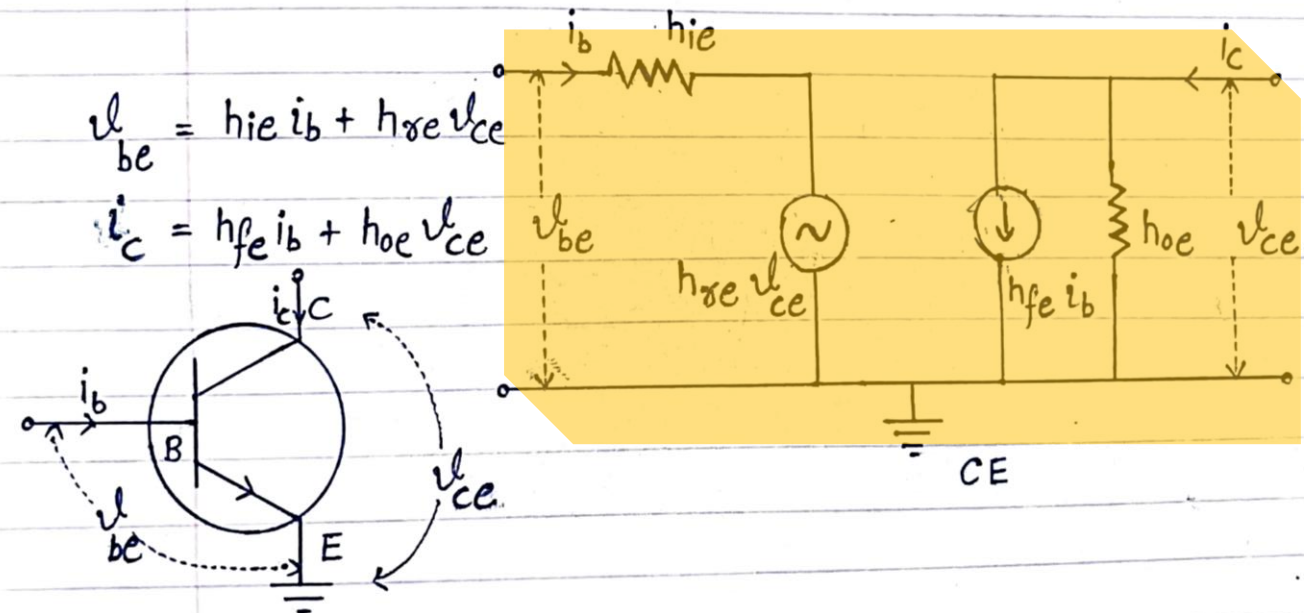
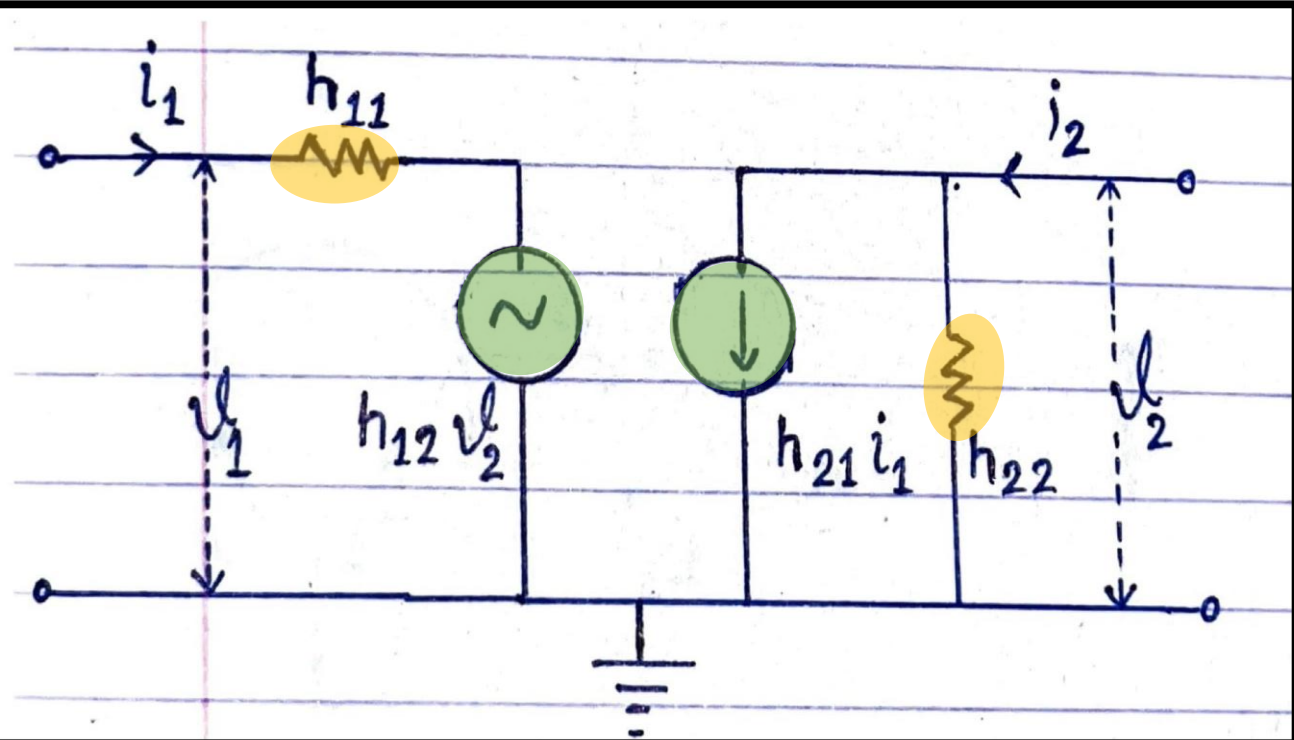
$$\frac{i_2}{v_2} = h_{22} = \text{o.c. output admittance, } i_1 = 0$$

$$= h_o; \text{ Unit} = \Omega (\text{mho})$$

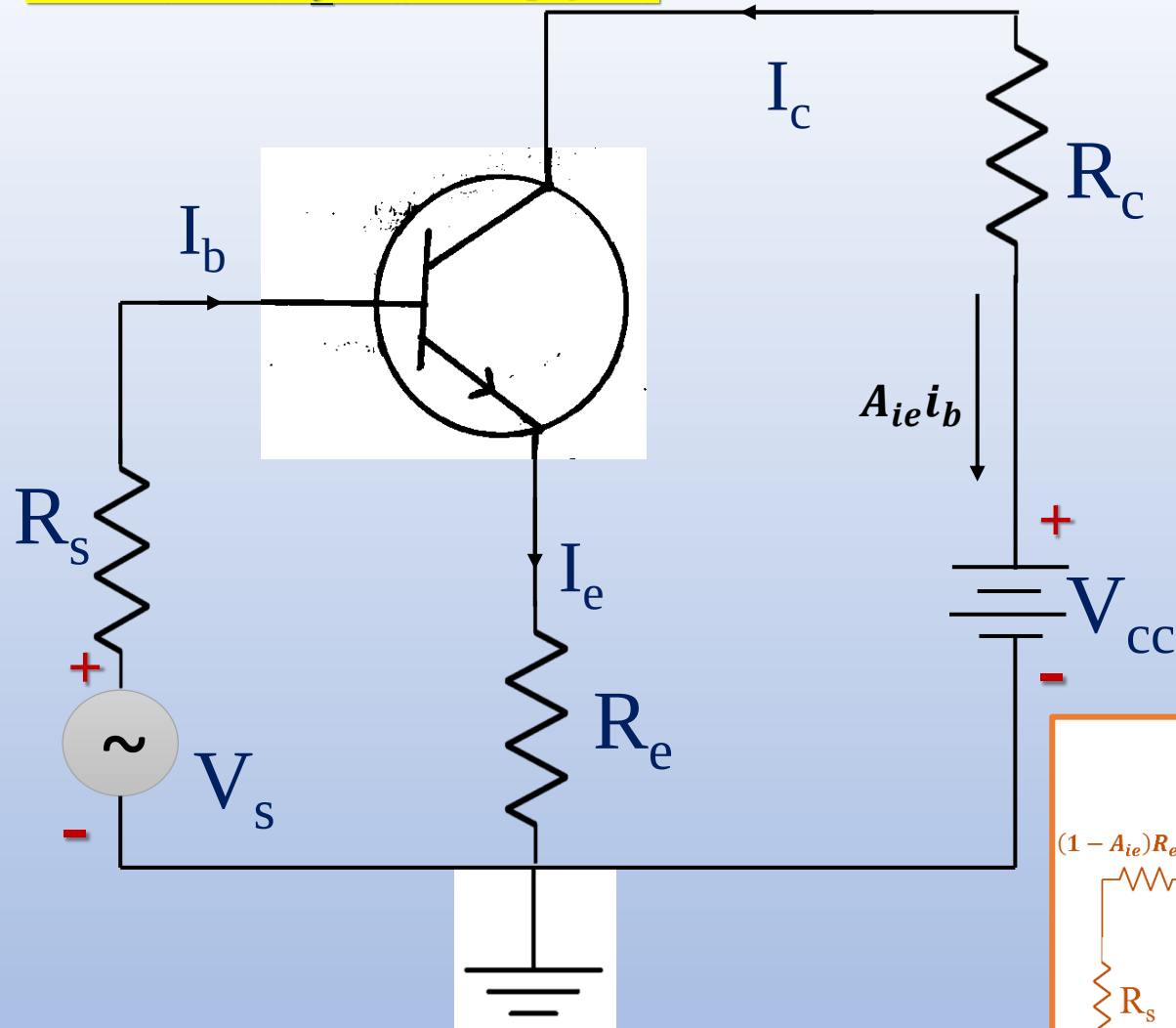
$$v_1 = h_{11} i_1 + h_{12} v_2$$

$$i_2 = h_{21} i_1 + h_{22} v_2$$

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \times \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$



■ Emitter અવરોધ R_e સાથે CE Amplifier -



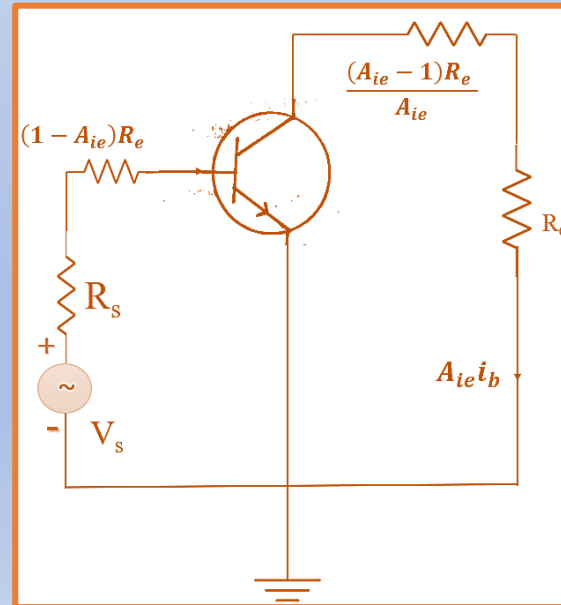
$$\text{Current Gain, } A_{ie} = - \frac{i_c}{i_b} \quad (1)$$

$$i_c = -A_{ie}i_b \quad (2)$$

$$i_e = i_b + i_c$$

$$i_e = i_b - A_{ie}i_b$$

$$i_e = i_b (1 - A_{ie}) \quad (3)$$



✓ R_e ને એમીટરથી દૂર કરી તેને input અને output ભાગમાં વિભાજિત કરતાં,

∴ $i_b R_b = i_e R_e$ માં $i_e = i_b (1 - A_{ie})$ મુકતાં,

$$R_b = (1 - A_{ie})R_e$$

$$i_c = -A_{ie}i_b \quad (\text{સમી. (2) પરથી})$$

$$i_c = - \frac{A_{ie}i_b}{(1 - A_{ie})} \quad (\text{સમી. (3) પરથી})$$

$$\therefore i_C = \frac{A_{ie} i_e}{(A_{ie} - 1)}$$

$$i_C R_C = i_e R_e \text{ માં } i_C = \frac{A_{ie} i_e}{(A_{ie} - 1)} \text{ મુકતાં,}$$

$$R_C = \frac{(A_{ie} - 1) R_e}{A_{ie}} \quad (4)$$

• પ્રવાહ લઘિય,

$$A_{ie} = -\frac{h_{fe}}{1 + h_{oe} R_L} \quad (5)$$

$$R_L = R_C + \frac{(A_{ie} - 1) R_e}{A_{ie}} \text{ સમી. (5) માં મૂકતાં,}$$

$$A_{ie} = -\frac{h_{fe}}{1 + h_{oe} \left(R_C + \frac{(A_{ie} - 1) R_e}{A_{ie}} \right)}$$

$$A_{ie} = -\frac{h_{fe} A_{ie}}{A_{ie} + h_{oe} R_C A_{ie} + (h_{oe} A_{ie} - h_{oe}) R_e}$$

$$-h_{fe} = A_{ie} + h_{oe} R_C A_{ie} + h_{oe} A_{ie} R_e - h_{oe} R_e$$

$$A_{ie} (1 + h_{oe} R_C + h_{oe} R_e) = h_{oe} R_e - h_{fe}$$

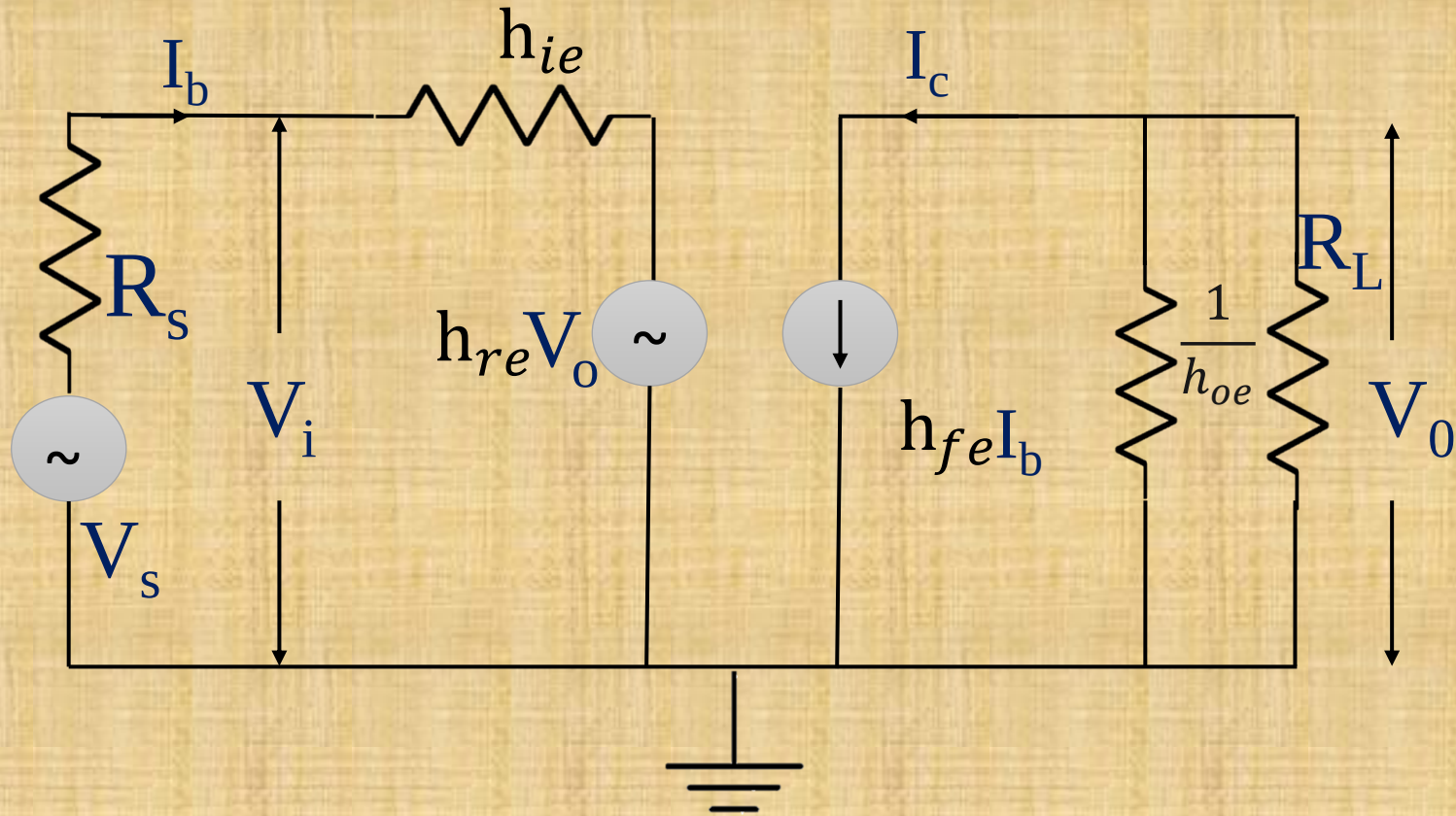
$$A_{ie} = \frac{h_{oe} R_e - h_{fe}}{(1 + h_{oe} (R_C + R_e))} \quad (6)$$

$$\text{વોલ્ટેજ લઘિય, } A_{Ve} = \frac{A_{ie} R_e}{R_{ie}} \quad (7)$$

$$\text{ઈનપુટ ઈમ્પેડન્સ, } R_{ie} = R_e (1 - A_{ie}) + h_{ie} + h_{ie} A_{ie} R_L$$

$$\text{આઉટપુટ એડમિટન્સ, } R_{oe} = h_{oe} + (R_s + h_{ie}) \left(1 + \frac{1}{h_{oe} R_e} \right)$$

CE હાઈબ્રીડ મોડેલનો સરળ h-પ્રાયલ પરિપથ -



$$\frac{1}{h_{oe}} \gg R_L$$

(Neglegable)

Forword Current Gain,

$$h_{fe} = - \frac{i_c}{i_b}$$

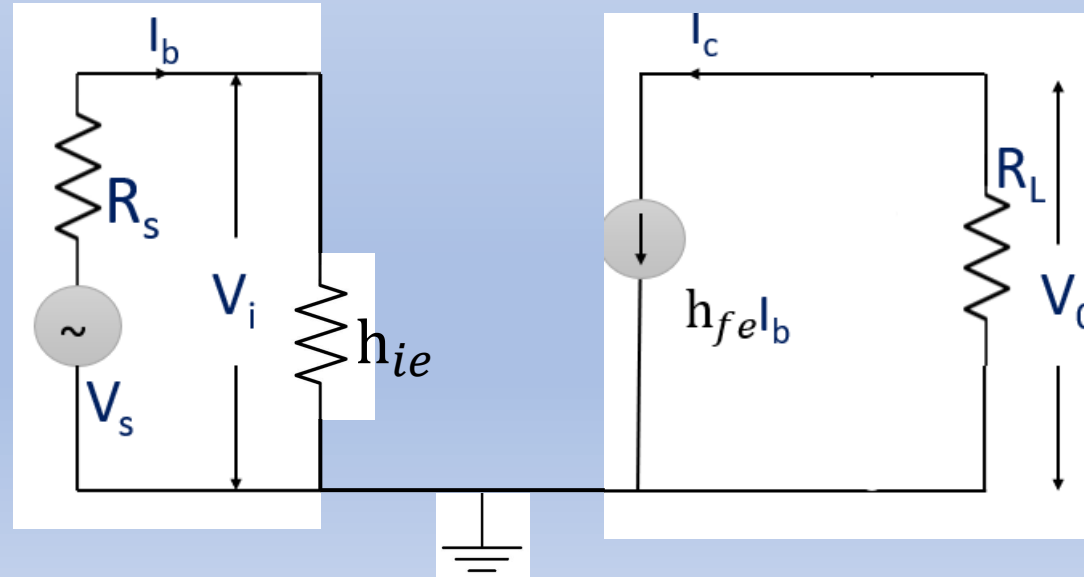
$$i_c = h_{fe} i_b \text{ — (1)}$$

- Generator Voltage, $h_{re}V_0 = h_{re}i_cR_L$

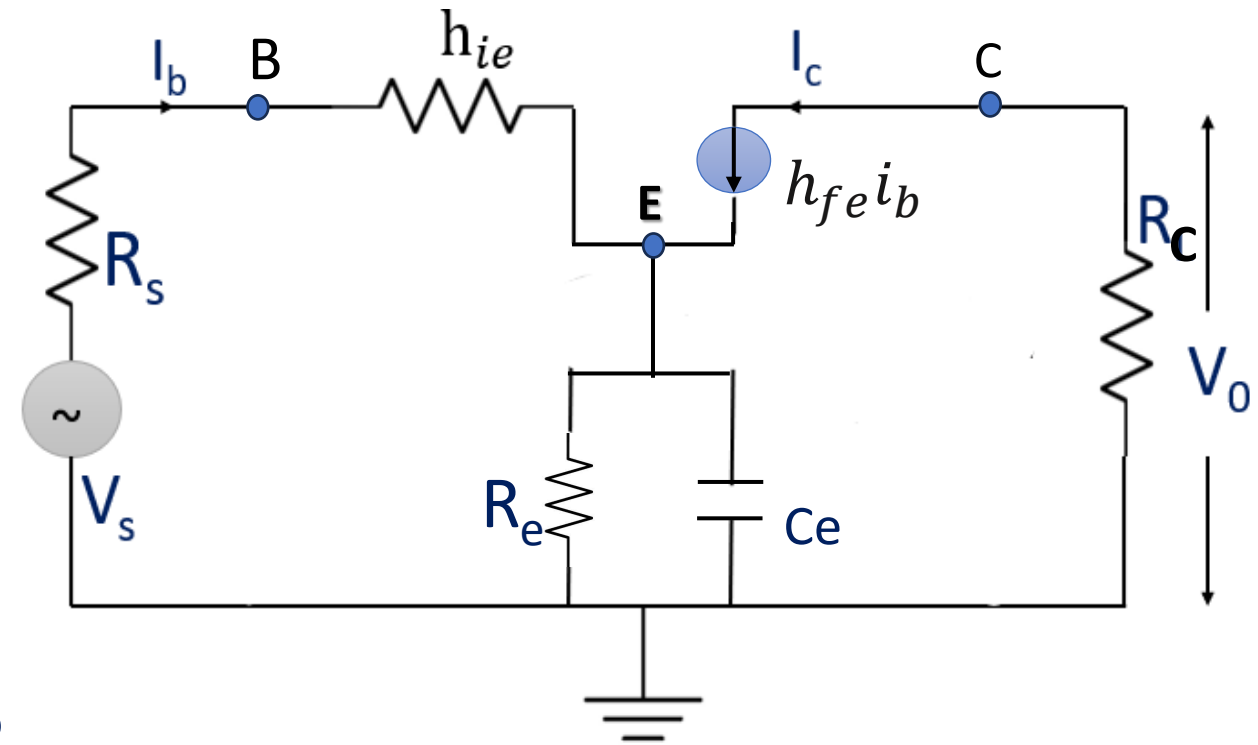
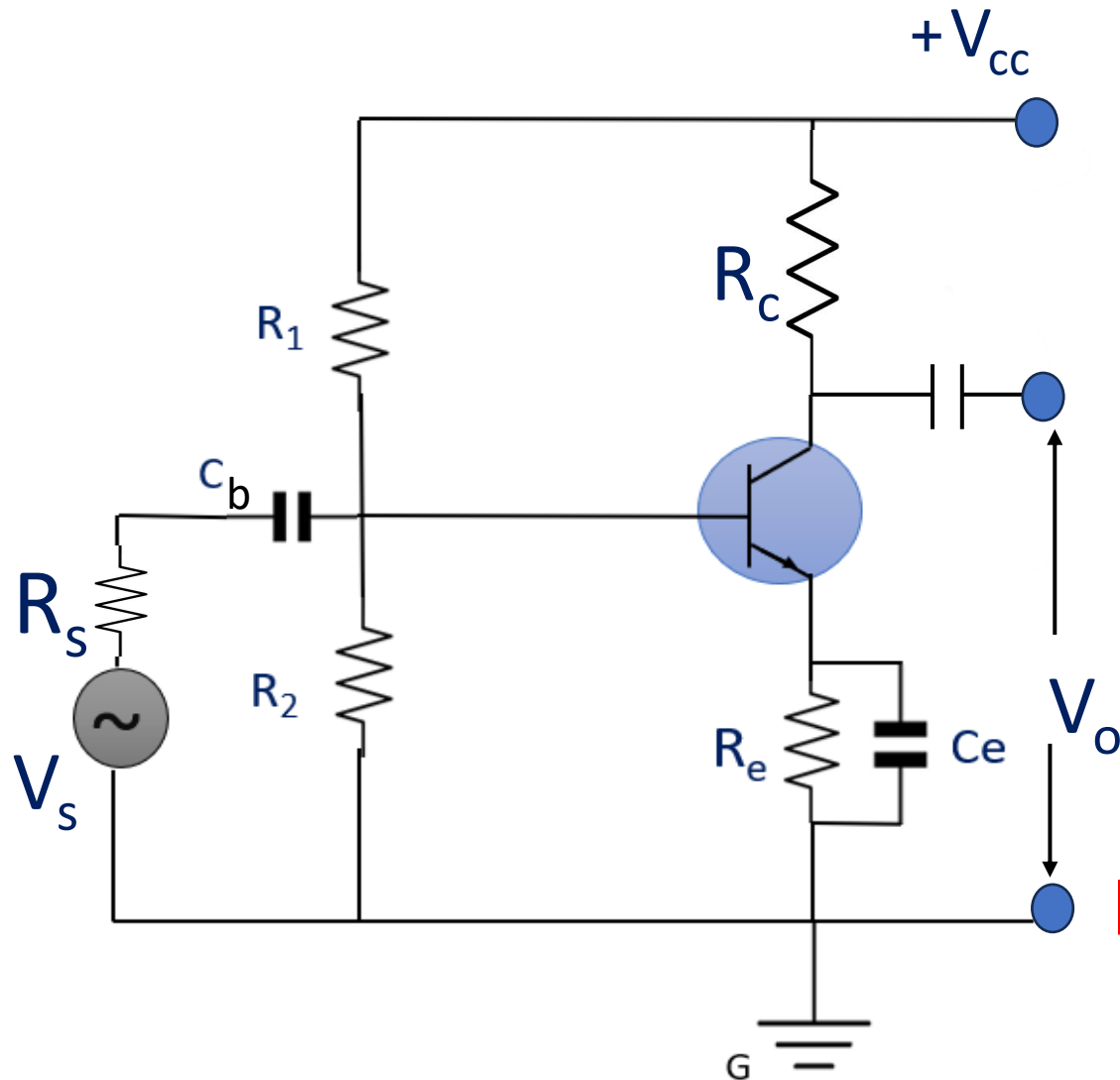
$$h_{re}V_0 = h_{re}h_{fe}i_bR_L$$

$$h_{re} \approx 0.01 \text{ (Neglegable)}$$

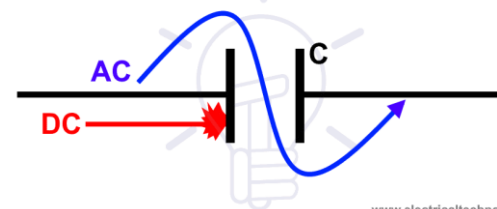
Simplification



Effect of Emitter Bypass Capacitor in Low Frequency Response -



Why Does A Capacitor Block DC But Pass AC?



DC = 0 Hz
AC = 50 Hz (India)

✓ Input Impedence,

$$h_{ie} = \frac{V_i}{I_i}$$

✓ Forword Current Gain,

$$h_{fe} = - \frac{i_c}{i_b}$$

✓ Output Voltage,

$$V_0 = - h_{fe} i_b R_c$$

(- Because of CE)

Base Current,, $i_b = \frac{V_s \text{ (Source Voltage)}}{R_s + R_i \text{ (Total Resistance)}}$

Input impedence, $R_i = h_{ie} + (1 + h_{fe}) Z_e$
(Unbypassed Circuit હોય તો Z_e ની અસર ઈનપુટમાં આવે.)

$$Z_e = \frac{R_e}{1 + j\omega R_e C_e} \text{ Because, } (R_e \parallel C_e)$$

Input impedence,

$$R_i = h_{ie} + (1 + h_{fe}) \frac{R_e}{1 + j\omega R_e C_e}$$

Base Current, $i_b = \frac{V_s}{R_s + h_{ie} + (1 + h_{fe}) \frac{R_e}{1 + j\omega R_e C_e}}$

✓ **Output Voltage,** $V_0 = -h_{fe} R_c \frac{V_s}{R_s + h_{ie} + (1 + h_{fe}) \frac{R_e}{1 + j\omega R_e C_e}}$

3

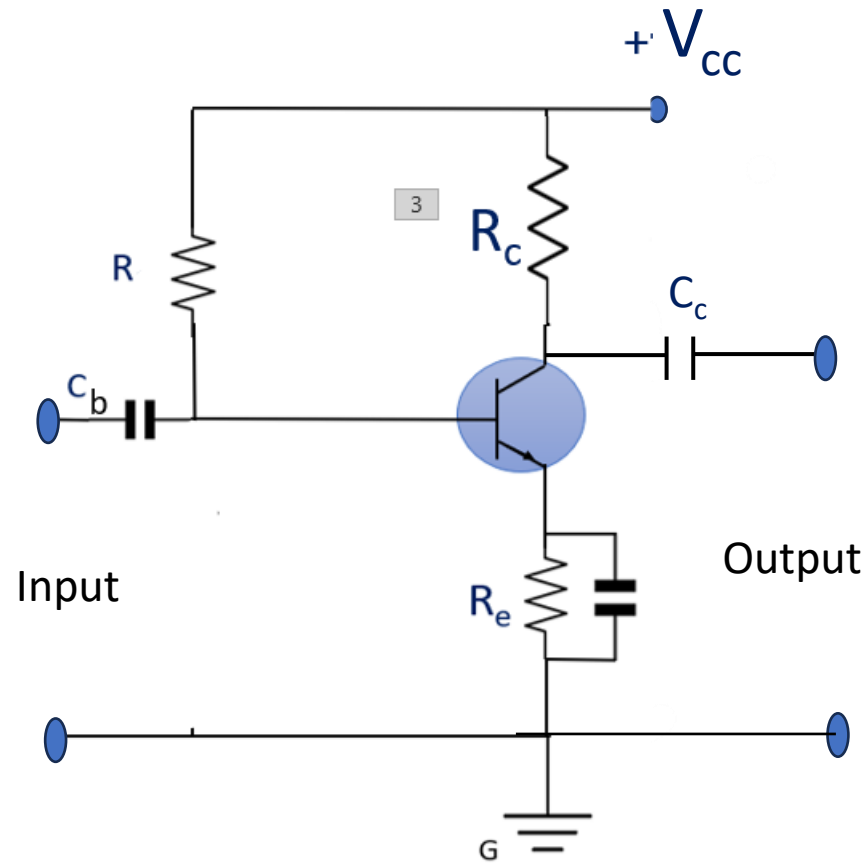
✓ Voltage Gain,

$$A_{ve}(L.F.) = \frac{V_0}{V_s} = - \frac{h_{fe} R_c}{R_s + h_{ie} + (1 + h_{fe}) \frac{R_e}{1 + j\omega R_e C_e}}$$

✓ Voltage Gain,

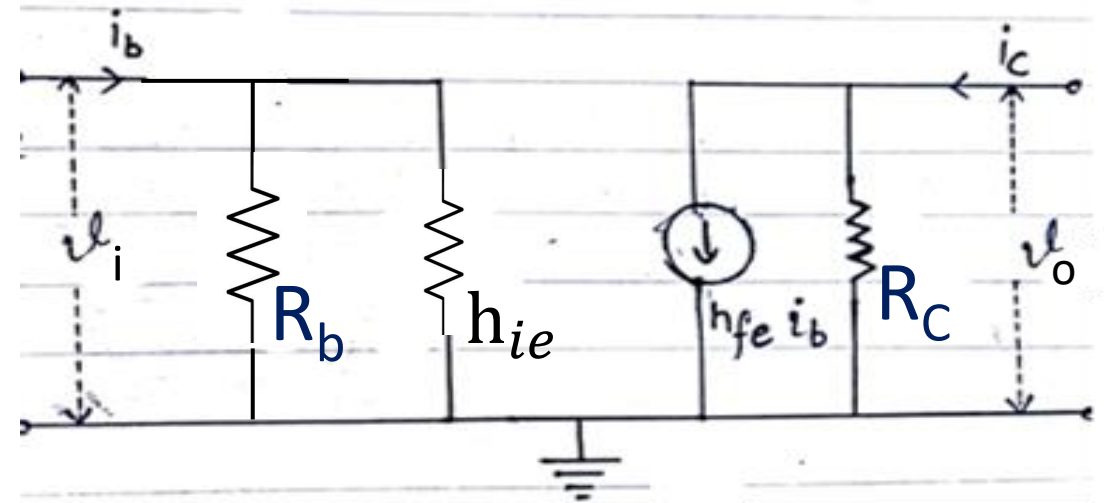
$$A_{ve}(M.F.) = \frac{V_0}{V_s} = - \frac{h_{fe} R_c}{R_s + h_{ie}}$$

■ Fixed Bias Method



1. *Input Impedence*, $R_{ie} = \frac{R_b h_{ie}}{R_b + h_{ie}}$

2. *Output Admittance*, $R_o = R_C$



3. *Current Gain*, $A_{ie} = h_{fe} = -\frac{i_c}{i_b} = \frac{h_{fe}}{1 + h_{oe} R_C}$ (With h_{oe})

4. Voltage Gain, $A_{ve} = \frac{V_o}{V_i}$

$$V_o = -i_c R_C = -h_{fe} i_b R_C \quad \& \quad V_i = h_{ie} i_b = i_b (h_{ie} \parallel R_b)$$

$$A_{ve} = -\frac{h_{fe}}{h_{ie}} R_C$$

$$A_{ve} = -\frac{h_{fe}}{h_{ie}} \left(R_C \parallel \frac{1}{h_{oe}} \right) \text{ (With } h_{oe} \text{)}$$

UNIT - II

(a) Basic Transistor Amplifiers:

Current and Voltage amplifiers (6.10), Common Emitter Amplifiers with Emitter Resistor (6.11), Simplified Common Emitter Hybrid Model (6.12), (6.12.1, 6.12.2), Effect of An Emitter Bypass Capacitor in low frequency Response (6.13)

(b) Multistage Amplifiers :

Multistage Transistor Amplifiers (General) (7.1), Terms used in Multistage Transistor Amplifiers (7.1.1), R-C- coupled Amplifiers (7.2), Middle Frequency Range (7.2a), Low Frequency Range and Lower Cut-off Frequency (7.2b) Transformer Coupled Amplifiers (7.4, a, b), Direct coupled Amplifiers (7.5), Effect of cascading on Band width (7.6).

Related Examples, Problems & Short Questions

Basic Reference : *Hand Book of Electronics by Gupta and Kumar. 46th revised Edition 2019.*

6.10 CURRENT AND VOLTAGE AMPLIFIERS

Voltage Amplifiers : The prime requirements for a voltage amplifier are :

- (a) When a signal generator is connected at its input, generator voltage should not change. For this condition to be satisfied input resistance (r_i) of the amplifier should be high;
- (b) Any change in load resistance should not affect the output voltage of the amplifier. This condition demands that output resistance of the amplifier (r_o) should be low.

For class A amplifiers, $r_i \cong \infty$ and $r_o < 1 \text{ k}\Omega$ and therefore this stage can be regarded as voltage amplifier.

Out of three configurations, only common collector amplifier has a high input resistance and a low output resistance and is therefore, operated as voltage amplifier.

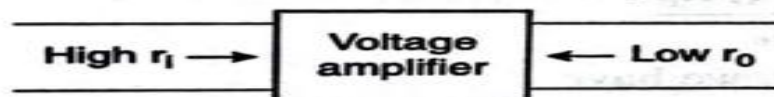


Fig. (31) Pictorial representation of a voltage amplifier conditions.



Fig. (32) Requirement of a current amplifier.

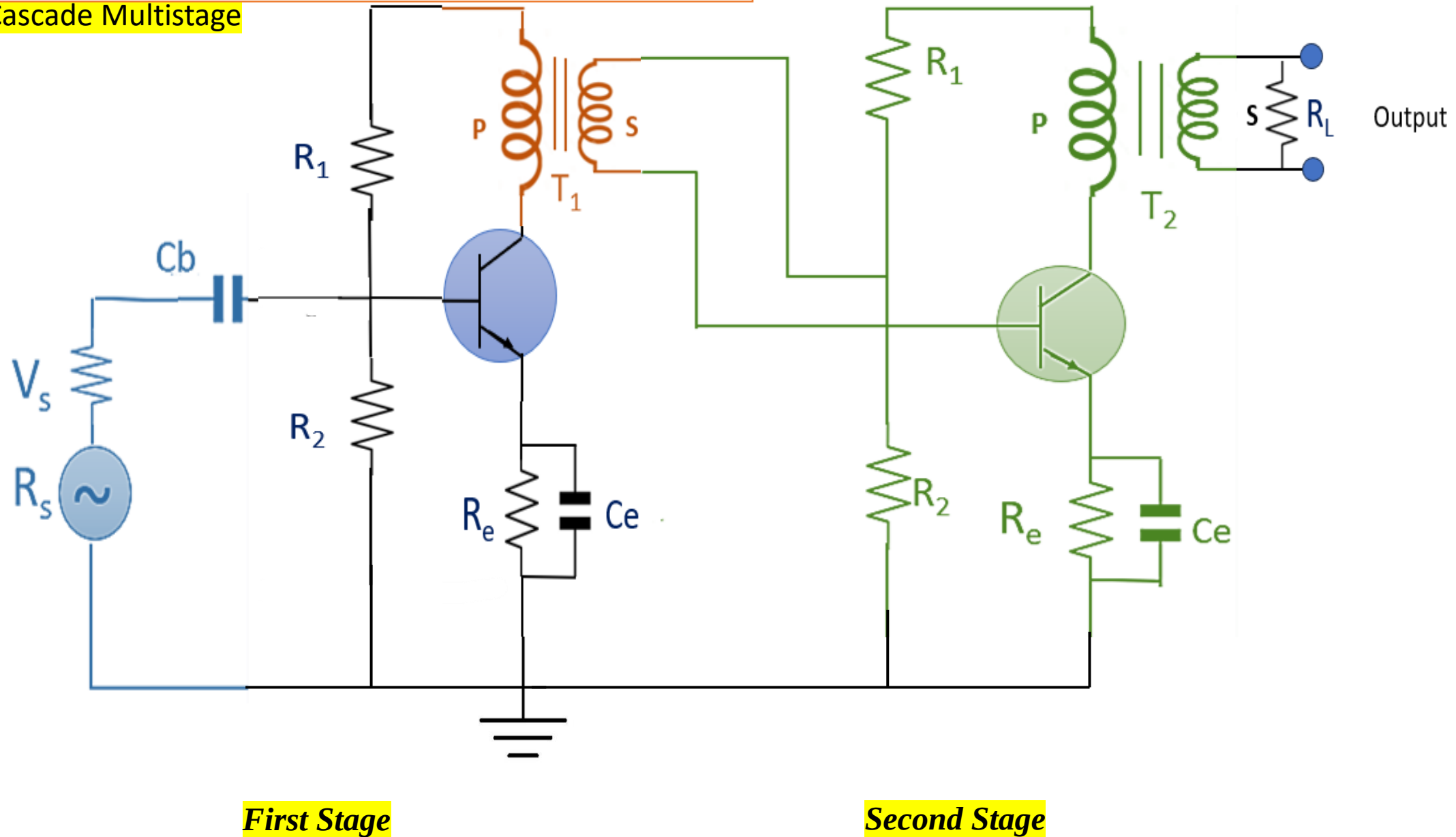
Current Amplifiers : The essential features of a current amplifier are :

- (a) when a signal generator is connected as its input, generator current should not change. This condition demands low value of input resistance (r_i).
- (b) any change in load resistance should not affect the output current, i.e., output current of the amplifier should be independent of load resistance. The condition demands a high value of output resistance (r_o) of the amplifier.

A common base and common emitter amplifier stage have small input resistance and high output resistance. Thus both configurations serve as current amplifier.

❖ Transformer Coupled Amplifier -

Cascade Multistage



✓ **Construction -**

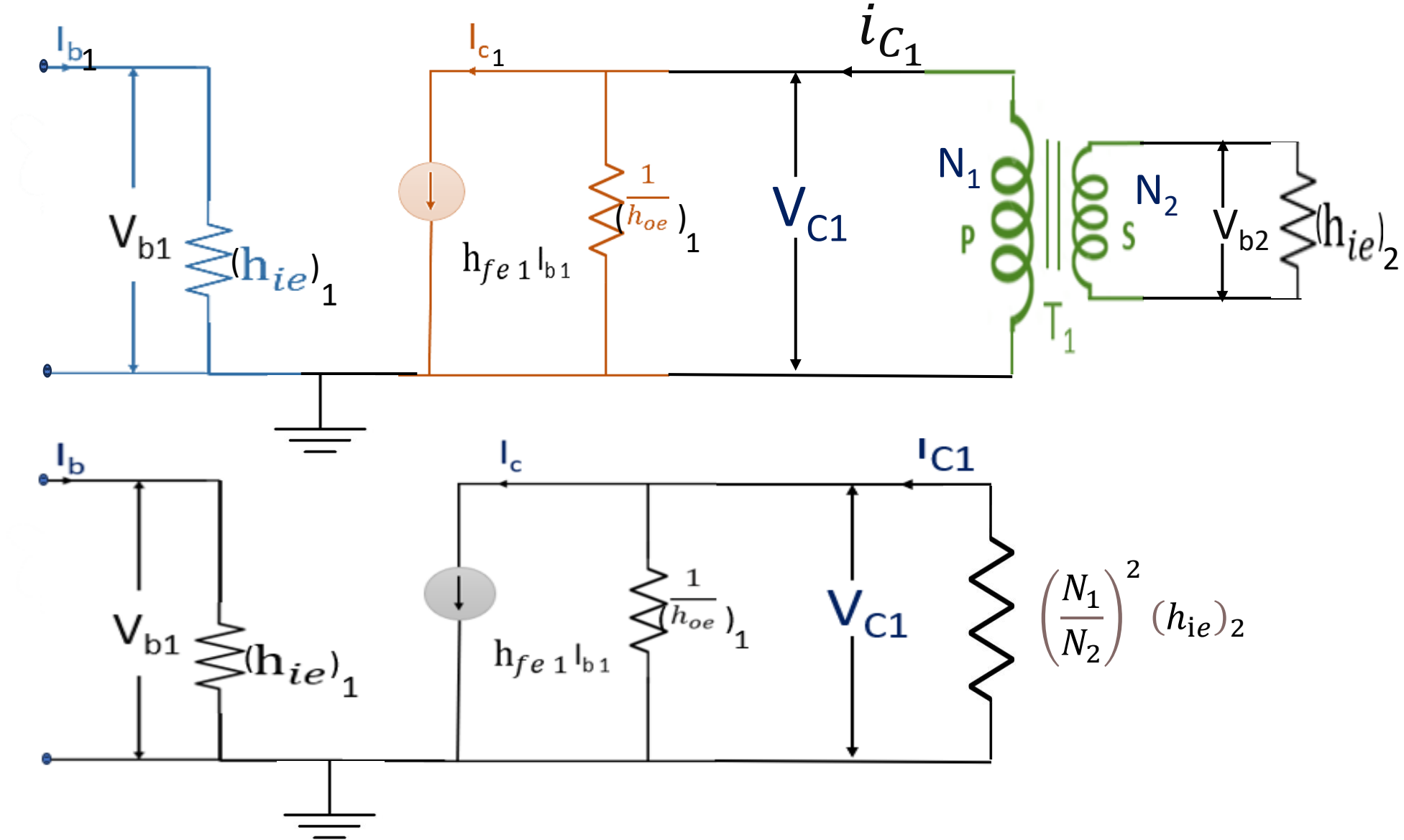
- ✓ The amplifier circuit in which, the previous stage is connected to the next stage using a coupling transformer, is called as Transformer coupled amplifier.
- ✓ The coupling transformer T_1 is used to feed the output of 1st stage to the input of 2nd stage.
- ✓ The collector load is replaced by the primary winding of the transformer. The secondary winding is connected between the potential divider and the base of 2nd stage, which provides the input to the 2nd stage.
- ✓ The potential divider network R_1 and R_2 and the resistor R_e together form the biasing and stabilization network.
- ✓ The emitter by-pass capacitor C_e offers a low reactance path to the signal. The resistor R_L is used as a load impedance.
- ✓ The input capacitor C_{in} present at the initial stage of the amplifier couples AC signal to the base of the transistor.

✓ **Working -**

- ✓ When an AC signal is applied to the input of the base of the first transistor then it gets amplified by the transistor and appears at the collector to which the primary of the transformer is connected.
- ✓ The voltage at the primary is transferred according to the turns ratio of the secondary winding of the transformer.
- ✓ This transformer coupling provides good impedance matching between the stages of amplifier because of stepdown transformer is used in output of second stage. The transformer coupled amplifier is generally used for power amplification.

✓ **Analysis -**

1. **Medium Range Frequency -**



✓ Output Impedance of Stage 1 = Input Impedance of Stage 2,

$$\frac{1}{(h_{oe})_1} = \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2$$

$$\left(\frac{N_1}{N_2}\right) = \frac{1}{\sqrt{(h_{oe})_1 (h_{ie})_2}} \quad \text{————— 1.}$$

✓ According to principle of Transformer,

$$\left(\frac{N_1}{N_2}\right) = \frac{i_{b_2}}{i_{c_1}} \quad \text{————— 2}$$

$$i_{b_2} = \left(\frac{N_1}{N_2}\right) i_{c_1}$$

$$i_c = h_{fe} i_b$$

▪ For Transformer, $i_c = \frac{h_{fe} i_b}{2}$

$$\text{so that, } i_{b_2} = \left(\frac{N_1}{N_2}\right) \frac{(h_{fe})_1 i_{b_1}}{2}$$

$$A_{ie} = \frac{i_{b_2}}{i_{b_1}} = \left(\frac{N_1}{N_2}\right) \frac{(h_{fe})_1}{2}$$

$$A_{ve} = \frac{v_{b_2}}{v_{b_1}}$$

$$v_{C_1} = -i_{C_1} \left(\frac{N_1}{N_2} \right)^2 (h_{ie})_2$$

- For Transformer, $i_c = \frac{h_{fe} i_b}{2}$

$$i_{C_1} = \frac{(h_{fe})_1}{2} \frac{v_{b_1}}{h_{ie_1}}$$

$$v_{C_1} = \frac{-(h_{fe})_1}{2} \frac{v_{b_1}}{(h_{ie})_1} \left(\frac{N_1}{N_2} \right)^2 (h_{ie})_2$$

$$\frac{v_{C_1}}{v_{b_1}} = \frac{-(h_{fe})_1}{2} \frac{1}{h_{ie_1}} \left(\frac{N_1}{N_2} \right)^2 (h_{ie})_2$$

- Multiply and divide by v_{b_1} ,

$$\frac{v_{C_1}}{v_{b_2}} \frac{v_{b_2}}{v_{b_1}} = \frac{-(h_{fe})_1}{2} \frac{1}{h_{ie_1}} \left(\frac{N_1}{N_2} \right)^2 (h_{ie})_2$$

$$\text{But, } \frac{v_{C_1}}{v_{b_2}} = -\frac{N_1}{N_2}$$

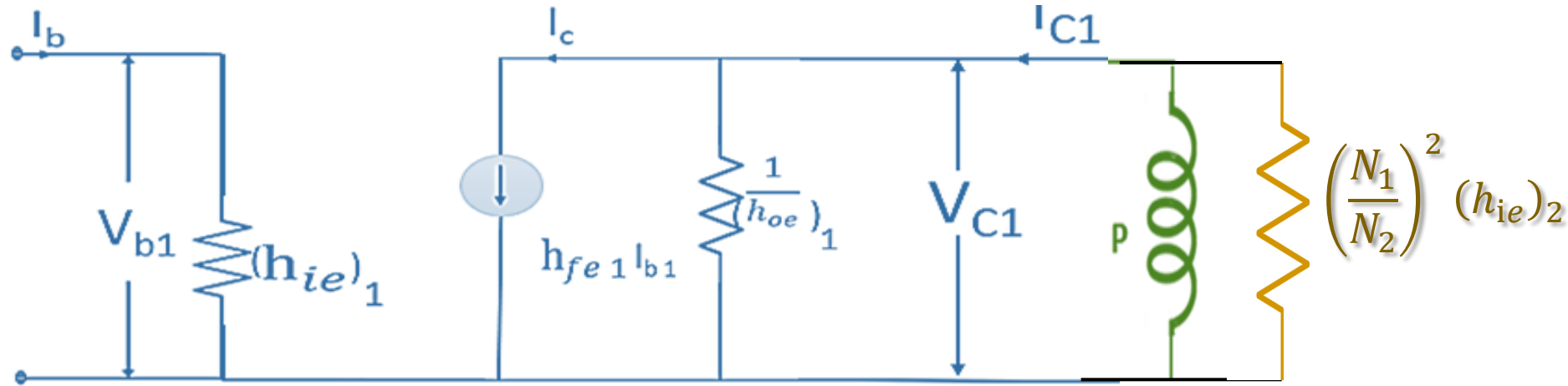
$$-\frac{N_1}{N_2} \frac{v_{b_2}}{v_{b_1}} = \frac{-(h_{fe})_1}{2} \frac{1}{h_{ie_1}} \left(\frac{N_1}{N_2} \right)^2 (h_{ie})_2$$

$$A_{ve} = \frac{v_{b_2}}{v_{b_1}} = \left(\frac{N_1}{N_2} \right) \frac{(h_{fe})_1}{2}$$

For Ideal Transformer, $h_{ie_1} = h_{ie_2}$

$$A_{ie} (\text{Mid.}) = A_{ve} (\text{Mid.})$$

2. Low Range Frequency – (Consider Effect of Primary Coil)



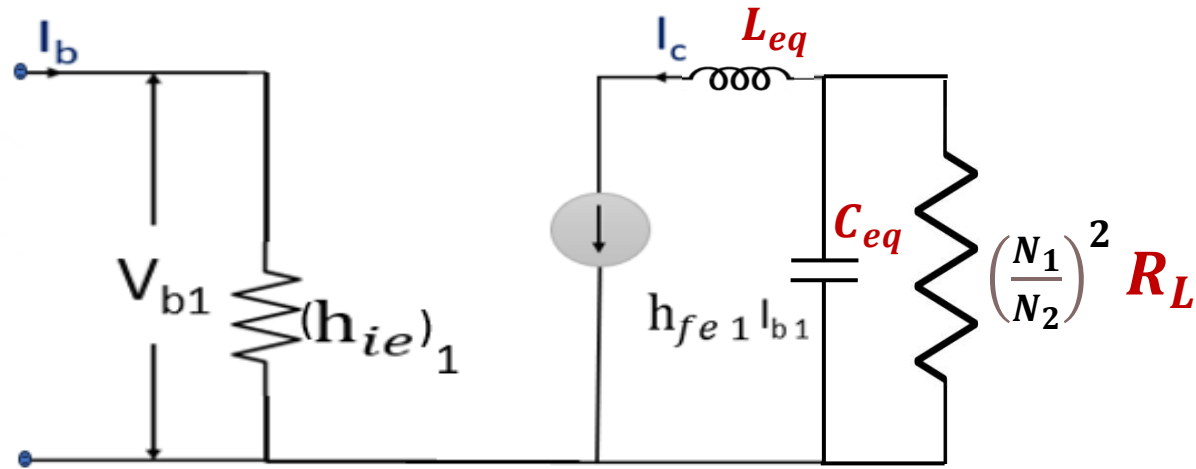
✓ *Output Admittance, $R = \frac{\frac{1}{h_{oe1}} \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2}{\frac{1}{h_{oe1}} + \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2}$*

✓ We know that, $X_L = 2\pi f_1 L_p$

✓ $f_1 = \frac{\left(\frac{1}{h_{oe1}} \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2\right)}{2\pi L_p \left(\frac{1}{h_{oe1}} + \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2\right)} = \frac{R}{2\pi L_p}$

✓ When the value of R and X_L are equal, $2\pi f_1 L_p = \frac{\frac{1}{h_{oe1}} \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2}{\frac{1}{h_{oe1}} + \left(\frac{N_1}{N_2}\right)^2 (h_{ie})_2}$

2. High Frequency Range – (medium between transformer's coil act as a dielectric material and transformer works as a capacitor)

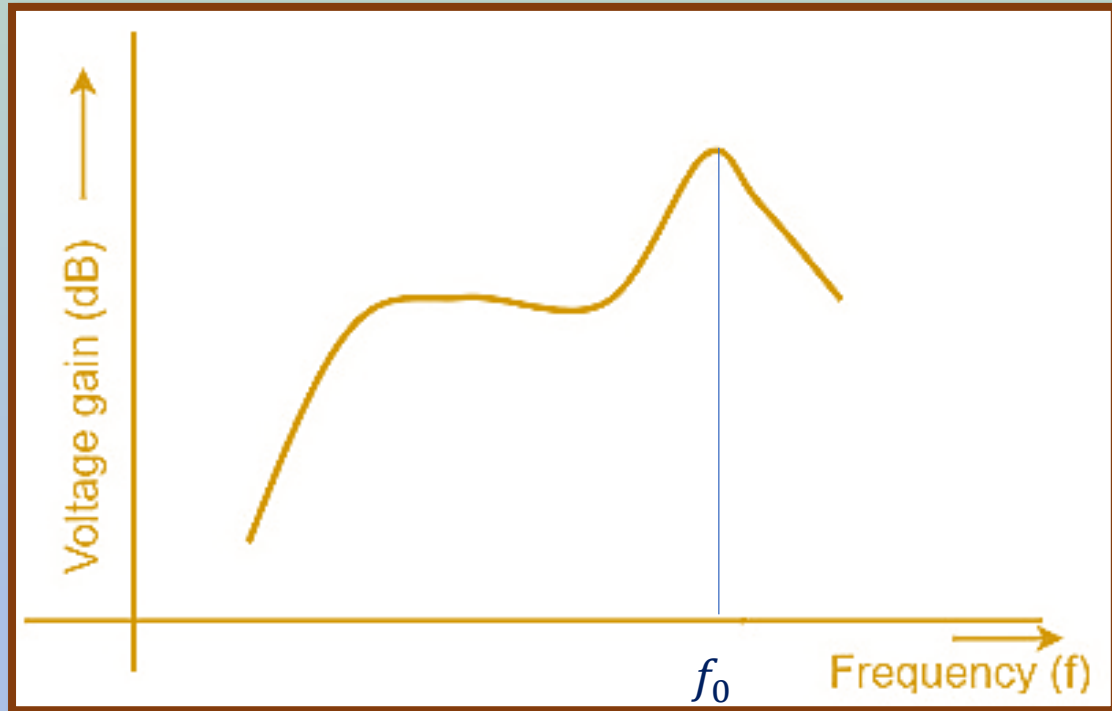


▪ Resonance Frequency, $f_0 = \frac{1}{2\pi\sqrt{L_{eq}C_{eq}}}$

$$L_{eq} = L_p + \left(\frac{N_1}{N_2}\right)^2 L_s \text{ and } C_{eq} = C_p + \left(\frac{N_1}{N_2}\right)^2 C_s$$

■ Frequency Response of Transformer Coupled Amplifier :

- ✓ the frequency response of an amplifier is a plot of gain v/s frequency.
- ✓ The figure below shows the frequency response of a transformer coupled amplifier. The gain of the amplifier is constant only for a small range of frequencies. frequency response of a transformer coupled amplifier is rather poor.
- ✓ The output voltage is equal to the collector current multiplied by the reactance of primary.



- At low frequencies, the reactance of primary begins to fall, resulting in decreased gain. At high frequencies, the capacitance between turns of windings acts as a bypass condenser to reduce the output voltage and hence gain.
- So, the amplification of audio signals will not be proportionate and some distortion will also get introduced, which is called as Frequency distortion.

➤ **Advantages of Transformer Coupled Amplifier :**

- An excellent impedance matching is provided.
- Gain achieved is higher.
- There will be no power loss in collector and base resistors.
- Efficient in operation.



➤ **Disadvantages of Transformer Coupled Amplifier :**

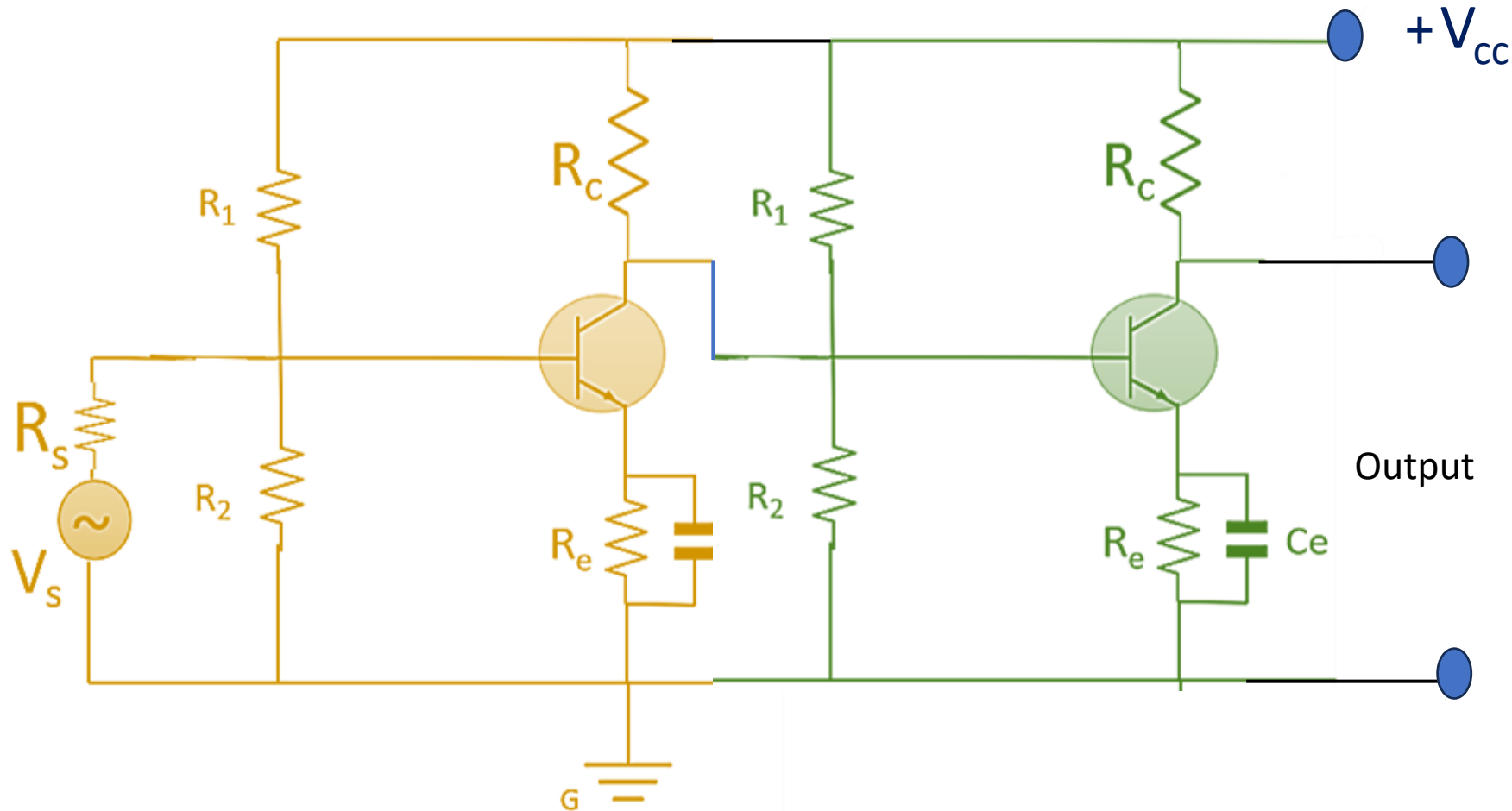
- Though the gain is high, it varies considerably with frequency. Hence a poor frequency response.
- Frequency distortion is higher.
- Transformers tend to produce hum noise.
- Transformers are bulky and costly.
- Working At Radio frequency(Higher frequency), it produce a lot of problems.



➤ **Applications :**

- costly used for impedance matching purposes.
- Used for Power amplification.
- Used in applications where maximum power transfer is needed.

▪ Direct Coupled Amplifier:



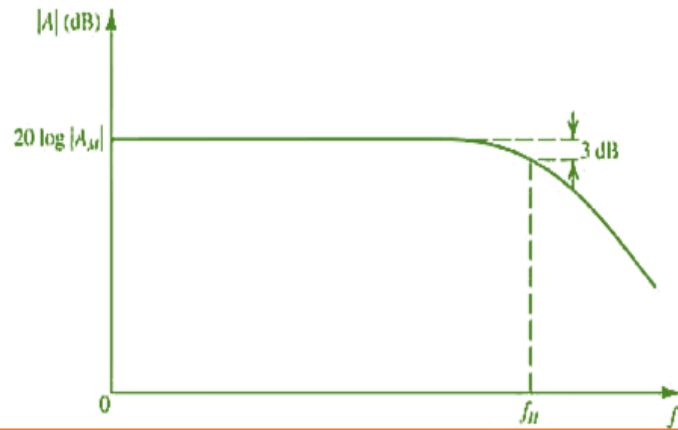
✓ Here, no Coupling is required so care has to be taken that the bias from the $+V_{cc}$ does not enter in output.

✓ So the biasing (Connections) becomes difficult in the circuit.

✓ And Total Gain is, multiplication of gain of both stages.

$$A_i = \beta_1 \beta_2$$

Frequency Response of DC Amplifier



✓ The figure below shows the frequency response of a Direct coupled amplifier. The gain of the amplifier is constant (Good response) only for a low frequency range.

➤ Advantages :

- The circuit arrangement is simple because of minimum use of ele. components.
- Good amplification efficiency at low frequency range.
- The circuit is of low cost because of the absence of expensive coupling devices.

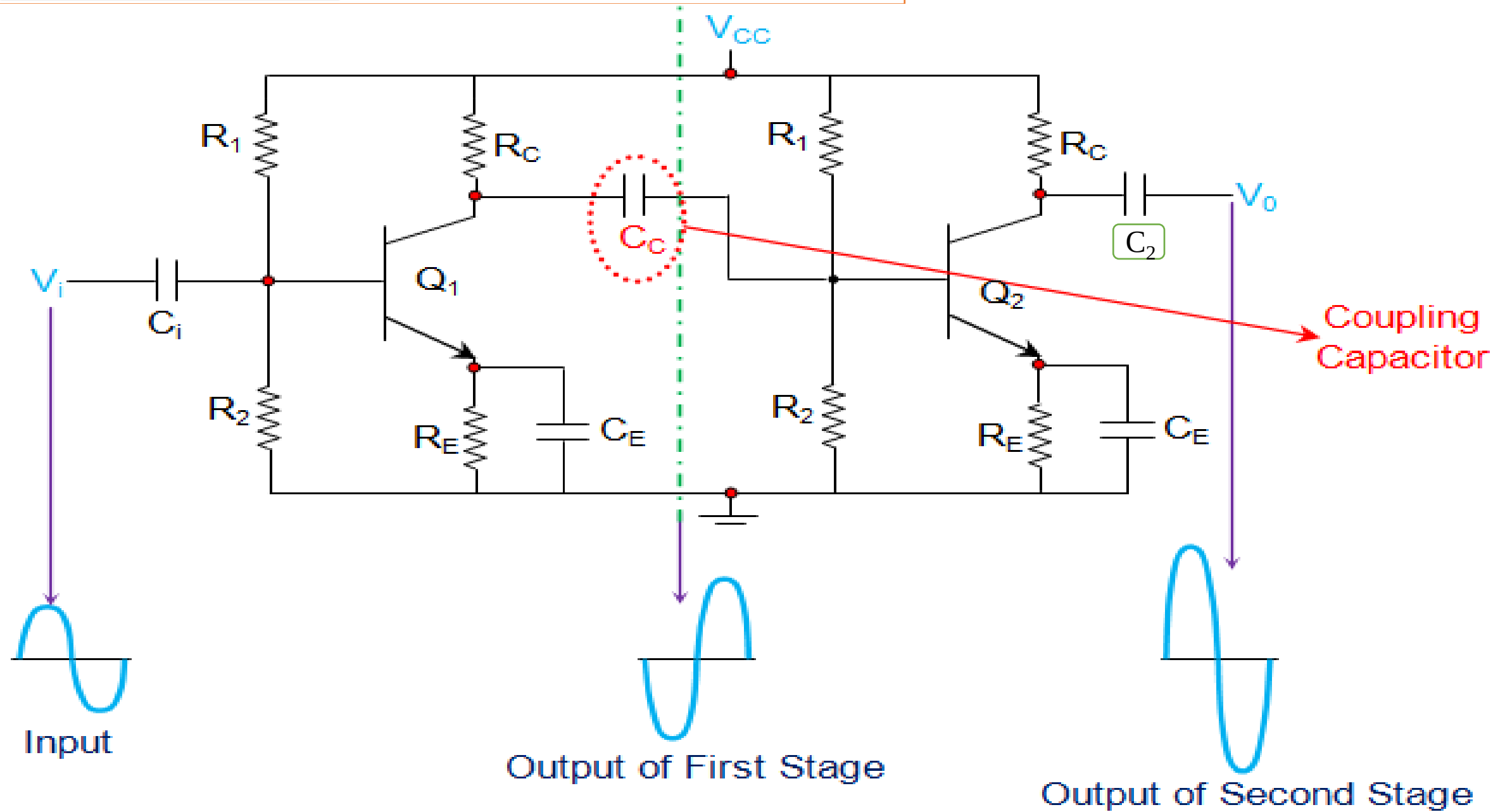
➤ Disadvantages :

- It cannot be used for amplifying high frequencies.
- The operating point is shifted due to temperature variations.

• Applications :

- Low frequency amplifications.
- Low current amplifications.

RC Coupled Amplifier.



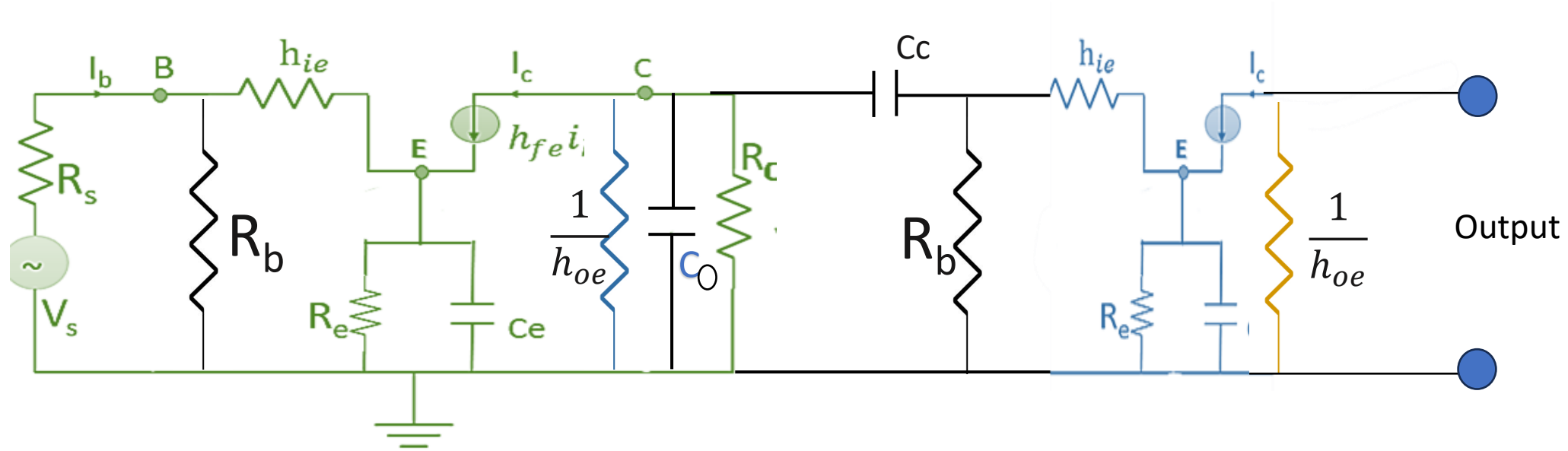
■ Construction -

- ✓ The two stage amplifier circuit has two transistors, connected in CE configuration and a common power supply V_{CC} is used. The potential divider network R_1 and R_2 and the resistor R_e form the biasing and stabilization network. The emitter by-pass capacitor C_e offers a low reactance path to the signal.
- ✓ The resistor R_L is used as a load impedance. The input capacitor C_{in} present at the initial stage of the amplifier couples AC signal to the base of the transistor. The capacitor C_C is the coupling capacitor that connects two stages and prevents DC interference between the stages and controls the shift of operating point.

■ Working -

- ✓ When an AC input signal is applied to the base of first transistor, it gets amplified and appears at the collector load R_L which is then passed through the coupling capacitor C_C to the next stage. This becomes the input of the next stage, whose amplified output again appears across its collector load. Thus the signal is amplified in stage by stage action.
- ✓ The important point that has to be noted here is that the total gain is less than the product of the gains of individual stages. This is because when a second stage is made to follow the first stage, the effective load resistance of the first stage is reduced due to the shunting effect of the input resistance of the second stage. Hence, in a multistage amplifier, only the gain of the last stage remains unchanged.
- ✓ As we consider a two stage amplifier here, the output phase is same as input. Because the phase reversal is done two times by the two stage CE configured amplifier circuit.

■ Analysis -

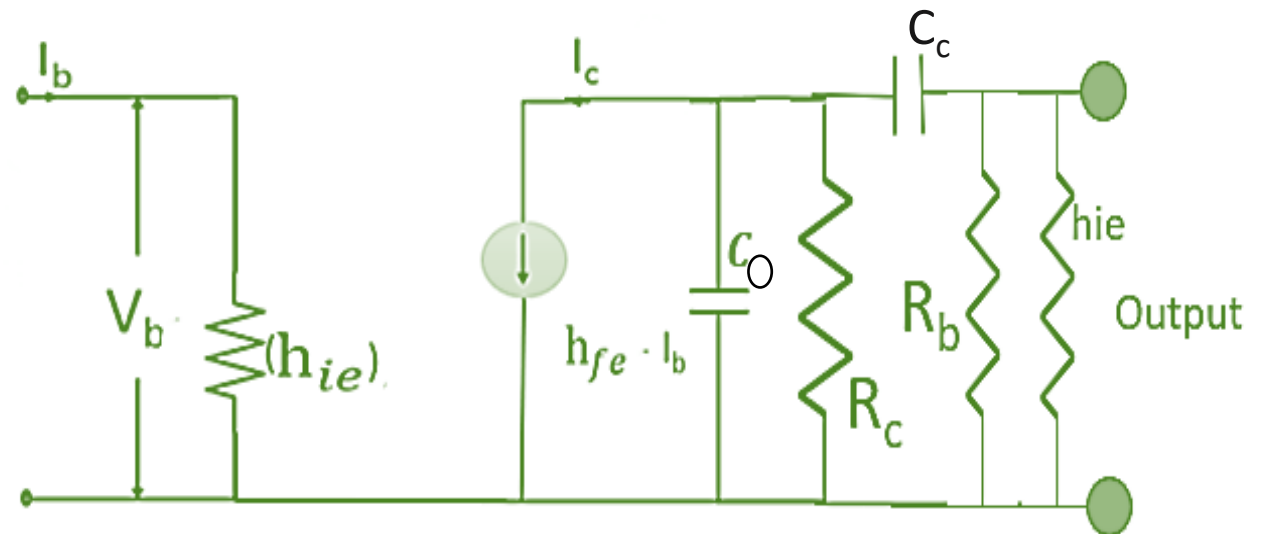


$$h_{re} \approx 0.01 \text{ (Neglegable)}$$

$$\frac{1}{h_{oe}} \gg R_L \text{ (Neglegable)}$$

$$R_e \parallel C_e \text{ (Neglegable)}$$

2nd stages o/p is depend on 1st stage i/p

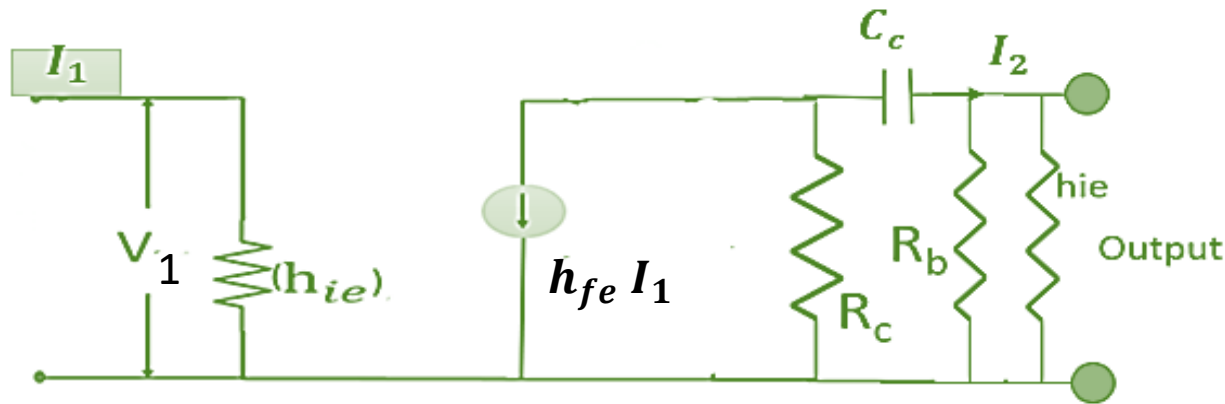


C_0 = Shunting Capacitor (output of 1st)

C_c = Coupling Capacitor

➤ **At Low frequencies (i.e. below 50 Hz) :**

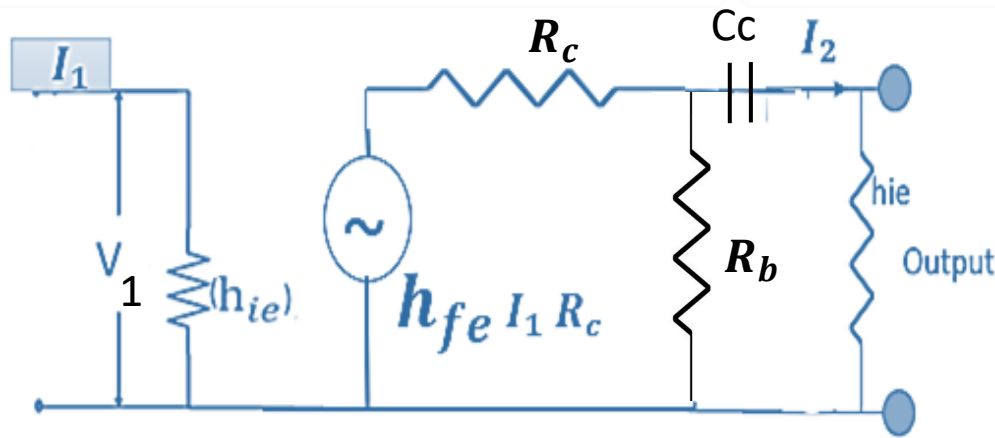
The capacitive reactance is inversely proportional to the frequency. At low frequencies, the reactance is quite high. The reactance of input capacitor C_{in} and the coupling capacitor C_c are so high that only small part of the input signal is allowed. The reactance of the emitter by pass capacitor C_E is also very high during low frequencies. Hence it cannot shunt the emitter resistance effectively. With all these factors, the voltage gain rolls off at low frequencies.



$$i_2 = \frac{\text{Voltage}}{\text{Resistance}} = \frac{h_{fe} I_1 R_c}{R_c + (R_b \parallel h_{ie}) + \text{Effect of } C_c}$$

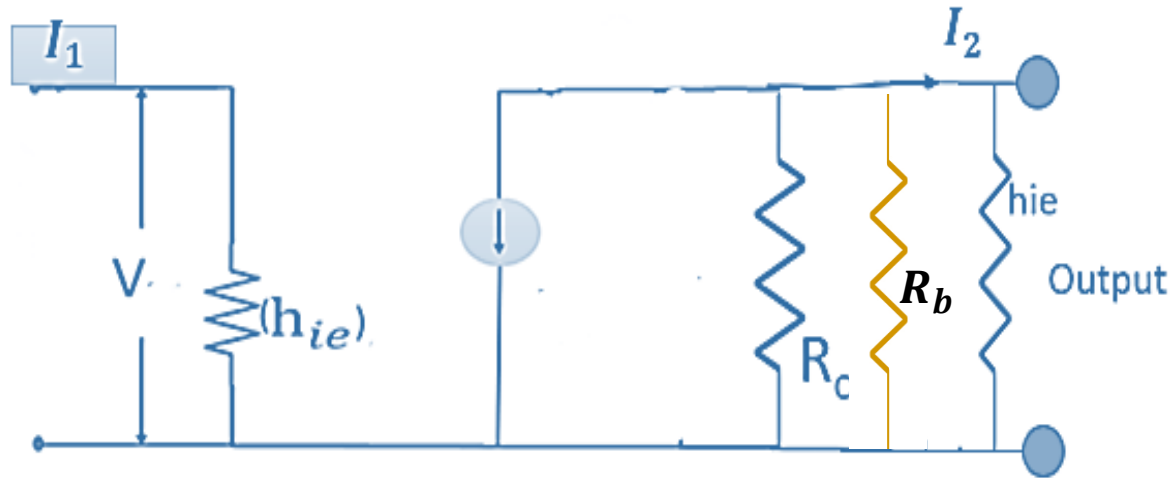
$$-\frac{i_2}{i_1} = A_{ie} (\text{Low}) = - \frac{h_{fe} R_c}{R_c + (R_b \parallel h_{ie}) + \text{Effect of } C_c}$$

C_o = Shunting Capacitor X
 C_c = Coupling Capacitor ☺

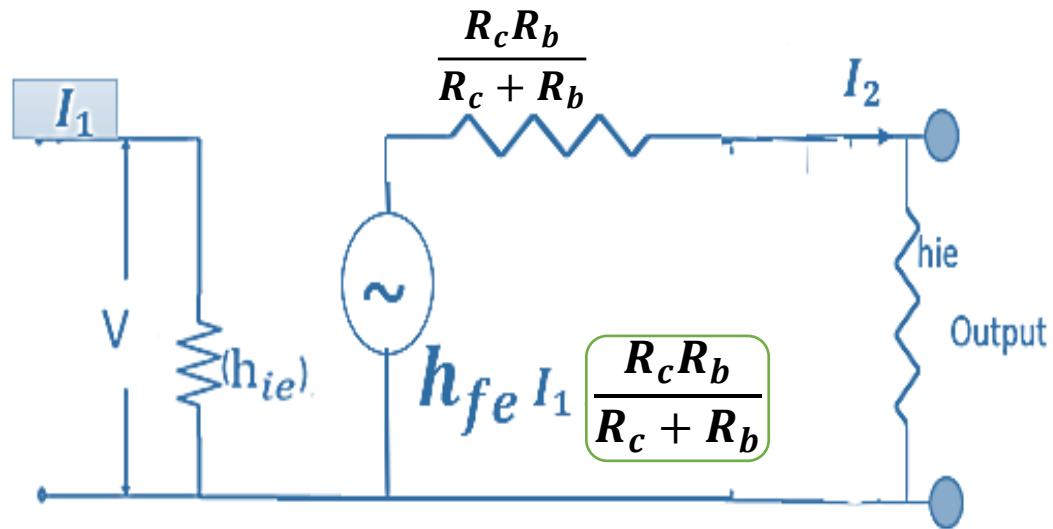


➤ At Mid-frequencies (i.e. 50 Hz to 20 KHz) :

C_0 = Shunting Capacitor X
 C_c = Coupling Capacitor X



$$i_2 = \frac{\text{Voltage}}{\text{Resistance}} = \frac{h_{fe} I_1 \left(\frac{R_c R_b}{R_c + R_b} \right)}{h_{ie} + \left(\frac{R_c R_b}{R_c + R_b} \right)}$$

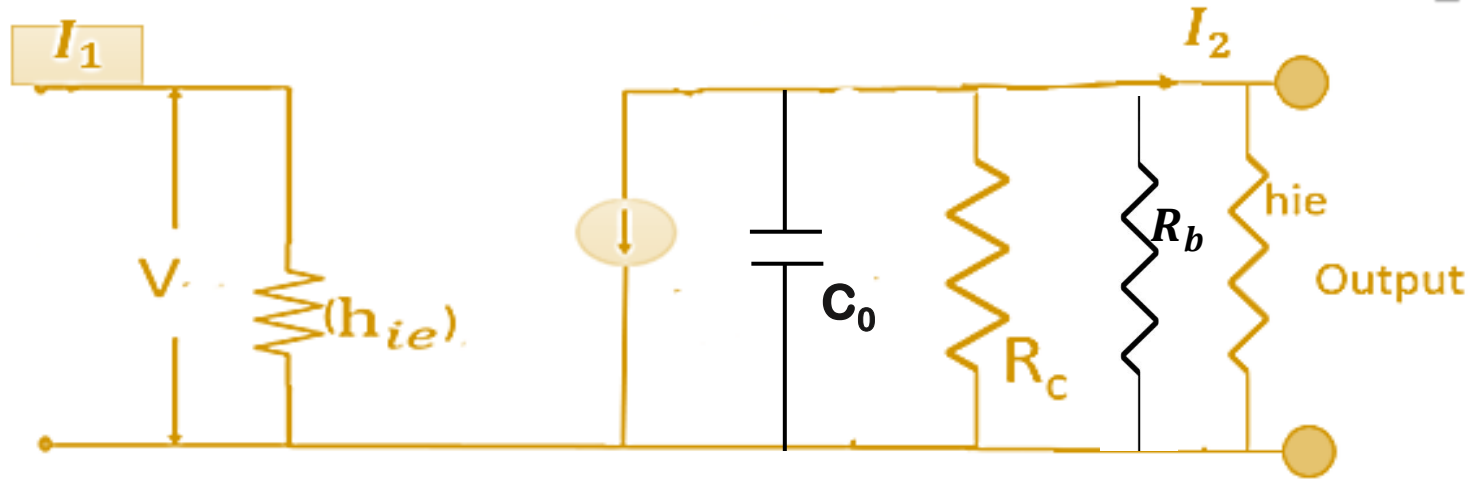


$$-\frac{i_2}{i_1} = A_{ie} (Mid) = -\frac{h_{fe} \left(\frac{R_c R_b}{R_c + R_b} \right)}{h_{ie} + \left(\frac{R_c R_b}{R_c + R_b} \right)}$$

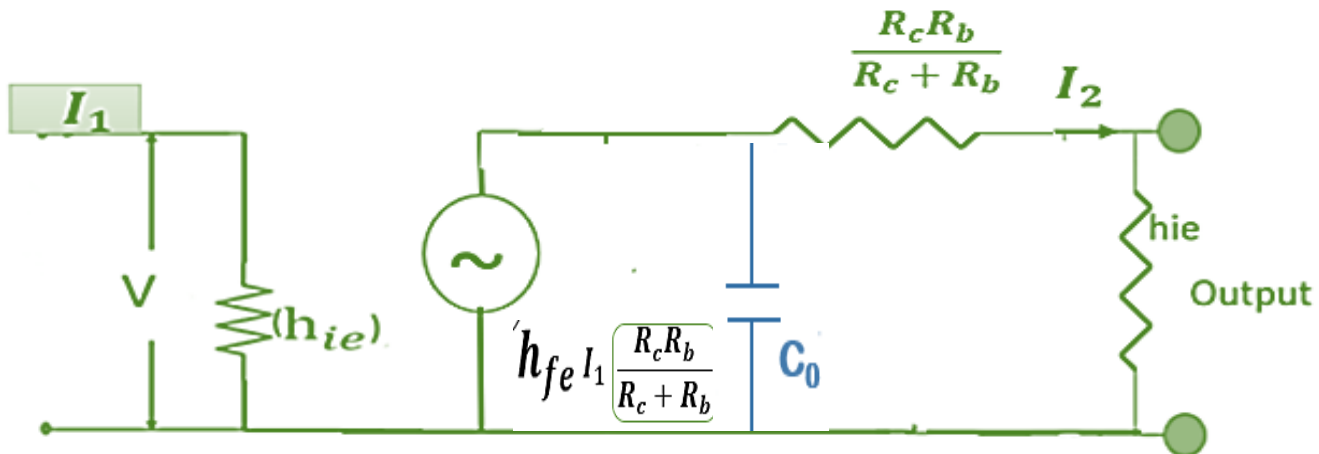
➤ **At High frequencies (i.e. above 20 KHz) :**

C_0 = Shunting Capacitor 😊

C_c = Coupling Capacitor ❌



$$i_2 = \frac{\text{Voltage}}{\text{Resistance}} \times \text{Effect of resistor to 2}^{\text{nd}} \text{ stage}$$

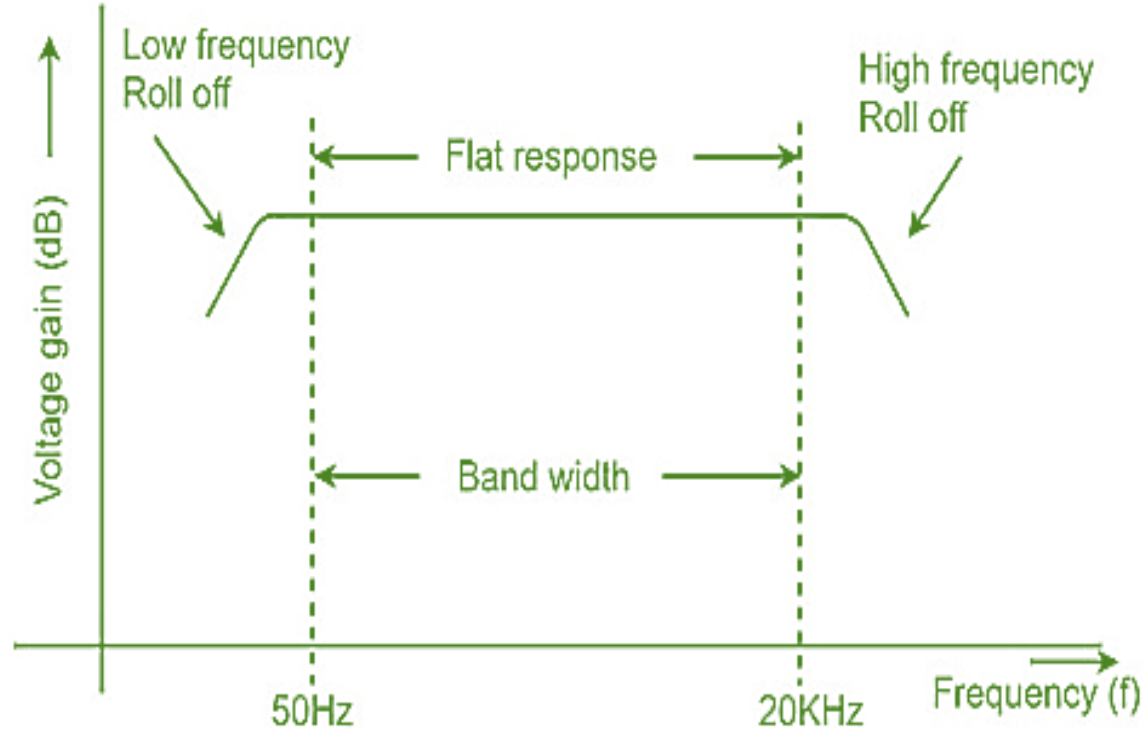


$$A_i (\text{Low}) = 0.707 I_{A_i} (\text{Mid})$$

$$A_i (\text{High}) = 0.707 I_{A_i} (\text{Mid})$$

■ Frequency Response of RC Coupled Amplifier :

- ✓ Frequency response curve is a graph that indicates the relationship between voltage gain and function of frequency. The frequency response of a RC coupled amplifier is as shown in the following graph.



- ✓ From the above graph, it is understood that the frequency rolls off or decreases for the frequencies below 50Hz and for the frequencies above 20 KHz. whereas the voltage gain for the range of frequencies between 50Hz and 20 KHz is constant.
- ✓ Good frequency response.
- ✓ Voltage gain constant at medium range.

➤ **Advantages of RC Coupled Amplifier :**

- Good frequency response. They have excellent audio fidelity over a wide range of frequency.
- Voltage gain constant at medium range.
- The circuit arrangement is simple.

➤ **Disadvantages of RC Coupled Amplifier :**

- They become noisy with age.
- Due to poor impedance matching, power transfer will be low.
- Due to poor impedance matching, RC coupling is rarely used in the final stages.

➤ **Applications of RC Coupled Amplifier :**

- Widely used as Voltage amplifiers

■ Questions -

- ✓ $\text{Gain} = \frac{\text{Output}}{\text{Input}}$ For Multistage $G = G_1 \times G_2 \times G_3 \dots$
- ✓ In many cases gain is denoted by bel or decibel (db) instead of numbers. For example logarithm of power gain is known as bel power gain.
- ✓ 10 x bel power gain = decibel Power gain.(1 bel = 10 db)
- ✓ Bandwidth = The difference between upper cutoff frequency (freq. at Max.gain x 0.707) and the lower cutoff frequency (freq. at Max. gain) is called bandwidth of frequency response graph.
- ✓ Need of Multistage ? The voltage or current gain obtained from single stage transistor amplifier is generally inadequate for most of the electronics applications.
- ✓ The output of each amplifier stage is coupled in some way to the input of the next stage.
- ✓ Need of Coupling ? To transfer the AC Output of 1st stage to input of the next stage. To prevent the DC from going to the next stage. Basic Methods are LS, RS, Transformer and Direct Coupling.
- ✓ Types of Multistage – 1. Cascade - Coupling and other components of each stage are the same.
2. Compound - Coupling and other components of each stage are not the same.

- **Effect of cascading on Bandwidth -**

$$(A_{Mid})_{total} = (A_{Mid})^n \quad \text{————— (1)}$$

$$A_{Low} = \frac{A_{Mid}}{1 - j\left(\frac{f_1}{f}\right)} \quad \text{————— (2)}$$

$$A_{Low (Total)} = \frac{A_{Mid (Total)}}{1 - j\left(\frac{f_{1n}}{f}\right)} \quad \text{————— (3)}$$

- From eqn. (1),

$$(A_{Low})_{total} = (A_{Low})^n$$

$$\frac{A_{Mid (Total)}}{1 - j\left(\frac{f_{1n}}{f}\right)} = \left(\frac{A_{Mid}}{1 - j\left(\frac{f_1}{f}\right)} \right)^n$$

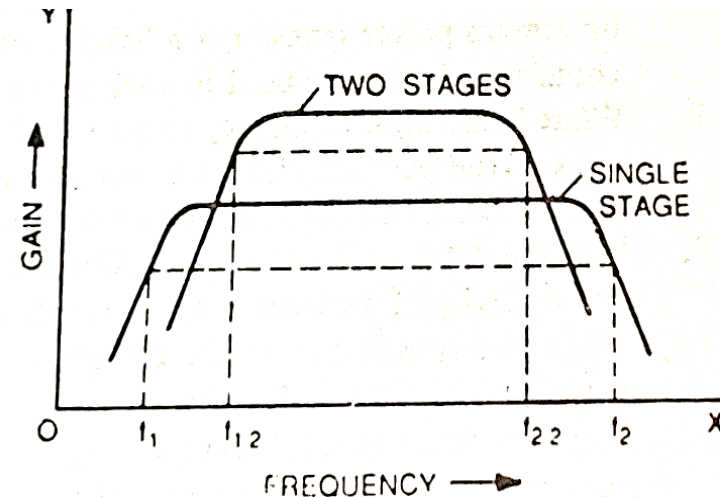
- From eqn. (1),

$$\frac{A_{Mid}^n}{1 - j\left(\frac{f_{1n}}{f}\right)} = \left(\frac{A_{Mid}}{1 - j\left(\frac{f_1}{f}\right)} \right)^n$$

$$1 - j\left(\frac{f_{1n}}{f}\right) = 1 - j\left(\frac{f_1}{f}\right)^n$$

- Equating the magnitude,

$$f_{1n} = \frac{f_1}{\sqrt{2^{\frac{1}{n}} - 1}}$$



○ Cascading.
Increment in gain
Decrement in bandwidth

S.No	Particular	RC Coupling	Transformer Coupling	Direct Coupling
1	Frequency response	Excellent in audio frequency range	Poor compare to RC Coupling	Best
2	Cost	Less	More	Least
3	Space and Weight	Less	More	Least
4	Impedance matching	Not good	Excellent	Good
5	Use	For voltage amplification	For Power amplification	For amplifying extremely low frequencies